

Book: The 8051 Microcontroller & Embedded Systems

Microprocessor Chip: 8051

First five chapters

Page 0-11

Slide ମଧ୍ୟମ୍ଭ ମଢ଼ାଣ

① Introduction to Computing

- Numbering System
- Digital Primer
- Inside a computer system

$\Rightarrow 25_{10}$ to $(25)_2$

$25/2 = 12$	1	↑	LSB
$12/2 = 6$	0		
$6/2 = 3$	0		
$3/2 = 1$	1		
$1/2 = 0$	1		MSB

$\Rightarrow (11001)_2$

$\Rightarrow (11001)_2$ to decimal

Base-16 Chart

A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z
.

Represent Binary to Hex

1001	1111	0101
9	F	5

Hex to Binary

9	F	5
↓	↓	↓
$(\quad)_2$		

Q1 Decimal to Hex

$\Rightarrow$ Decimal to Binary $\Rightarrow (1001\ 1001)_2$

$\Rightarrow$ Binary to Hex $\Rightarrow (1001\ 1001)_2$
 $\downarrow\quad\quad\downarrow$
 $(\quad\quad)_{16}$

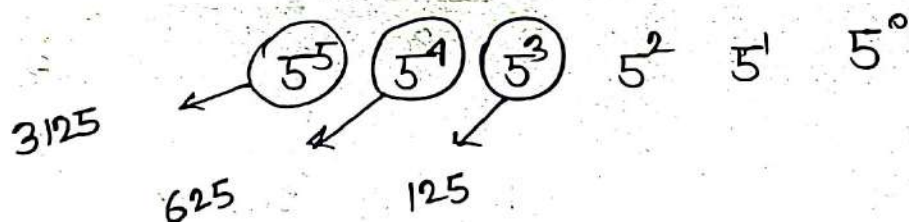
Q2 Hex to Binary then Binary to Decimal

Convert from base 5 to base 3

$$(4231)_5 = (?)_3$$

4 2 3 1

$5^5\ 5^4\ 5^3\ 5^2\ 5^1\ 5^0$



$$(4321)_5$$

$$\hookrightarrow 4 \ 3 \ 2 \ 1$$

$$4 \times 5^3 + 3 \times 5^2 + 2 \times 5^1 + 1 \times 5^0$$

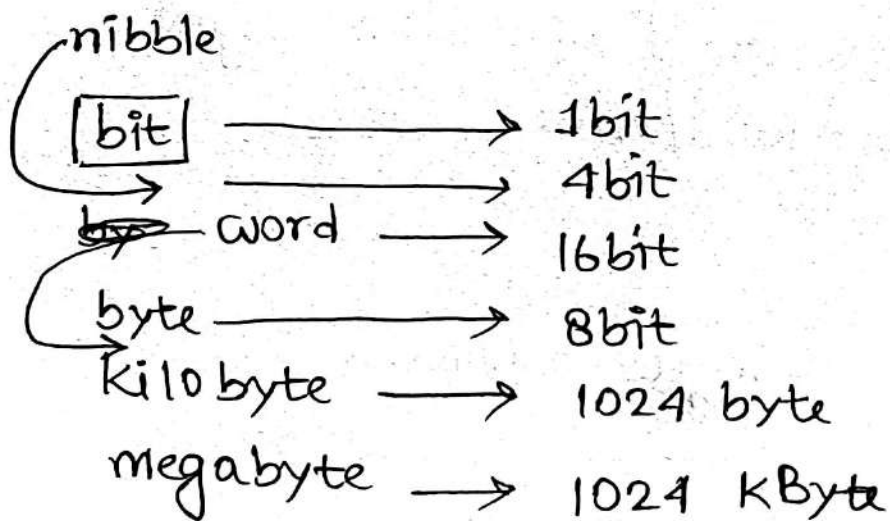
$$\Rightarrow (566)_5$$

$$566/3 \rightarrow$$

Hex

A(41)	a(61)
-------	-------

Rearrange the size from smaller to bigger.



What is inside a CPU?

CPU

→ Execute information stored in memory

I/O

→ Provide means of communicating with CPU

Memory

→ Types of Memory

Non-volatile

↳ Persistent Data

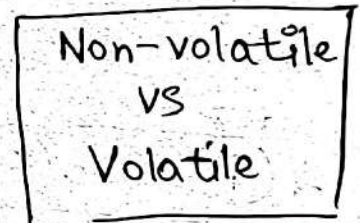
Volatile Data

Page 26 // Buses

⇒ Address bus

⇒ Data bus

⇒ Control Bus



Control bus → Read/Write operation signal

Data bus → Read/Write

Address bus

↳ Assign unique addresses

Databuses are bidirectional

↑ bus count ↑ speed

More data buses mean expensive CPU & computer

↳ Max size 8-64 & 128

$$\text{Address buses}^{\text{lines}} = 24$$

$$\text{Addressable Memory} = 2^{24}$$

Address buses are unidirectional.

⇒ Secondary & Primary Memory Difference



↗ Registers

⇒ Store temporary data/info

⇒ More register ↑ More cost ↑

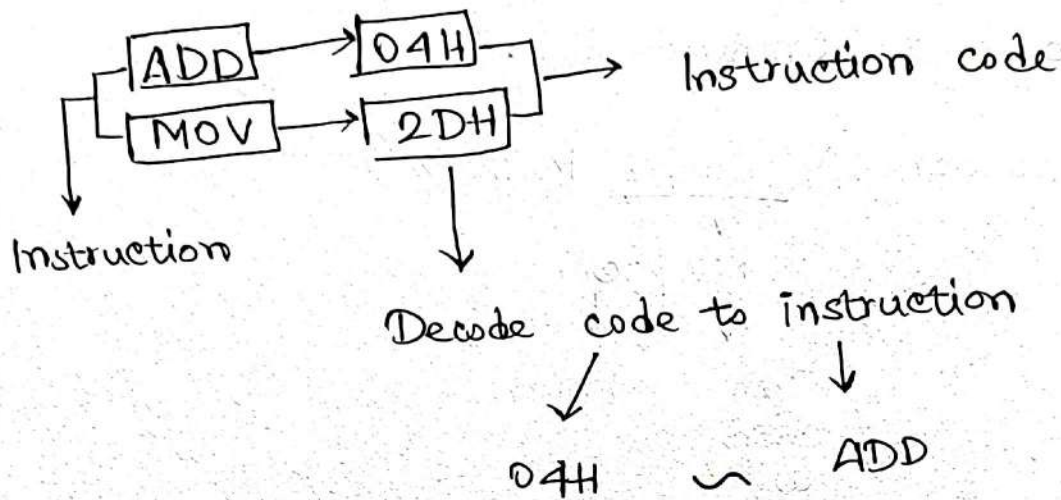
↗ Inside the CPU



figure: from PDF

➔ Instruction Decoder

Interprets the instruction fetched into the CPU.



⇒ PC, ALU, Instruction Decoder

➔ Internal Working of Computer (page 34)

Example

Page 34 -

A, B, C, D → 8 bit data bus

16 bit → address bus

CPU can access 0000 to FFFFH

Answer format
↓
Explanation

MOV A, 21H

ADD A, #42H

ADD A, #12H

ADD, MOV op → Program Counter 2 बार चलेगा

Register op → " " 1 बार चलेगा

ayan

MDV, ADD $\Rightarrow$ 2 val $\Rightarrow$ PC++2;

MOV, ADD $\Rightarrow$ single val $\Rightarrow$ PC++;

SUMMARY

- $\rightarrow$ Number Conversion
- $\rightarrow$ RAM, ROM, PM, SM, ALU, CPU, I/O
- $\rightarrow$ Internal Working of CPU

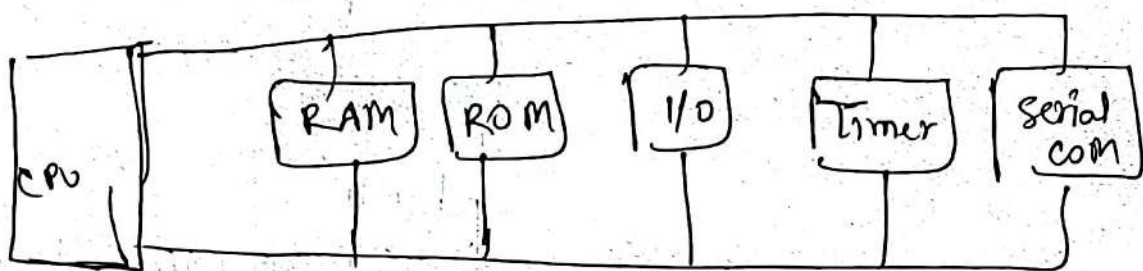
Assignment (Mid)

(Handwritten)

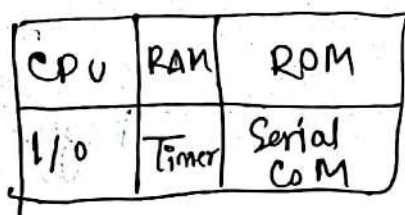
$\rightarrow$ Difference between Microcontroller & Microprocessor

Microcontroller $\rightarrow$ 1টি বক্স
(Projector, AC, TV) | Price low

Microprocessor $\rightarrow$ অনেক বক্স
(PC) | Price high



Microprocessor



Microcontroller

① General Purpose Microprocessor

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② Microcontroller

③ Embedded product

≠ embedded processor ~ microcontroller

≠ critical need = less power consumption & space

Chapter II

① Microprocessor vs Microcontroller

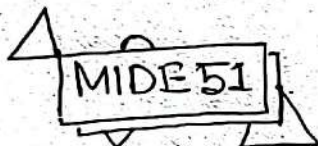
② Criteria for choosing a micro controller

③ Description of 8051 with figure (page 52, 53, 66)

PDF

Interrupt → doing quick favors

≠ Chapter III ← inside the 8051



DPTR

(Data pointer - 16 bit)

PC (16bit)

DPH

↳ Data pointer high

DPL

↳ Data pointer low

Register → 8bit

MOV A, #50H

↑
value denote করতে
দেওয়া must

না দিলে memory location বুঝায়

#OYSHI → Name

317CR → location

'0' at the front means it is a hex

& not a letter → #0F9

→ #05

Illegal: 7F2H > 8 bits (FFH)

ADD

(A)

Value / memory / register

Accumulator
& must

MOV R3, #35H
ADD A, R2

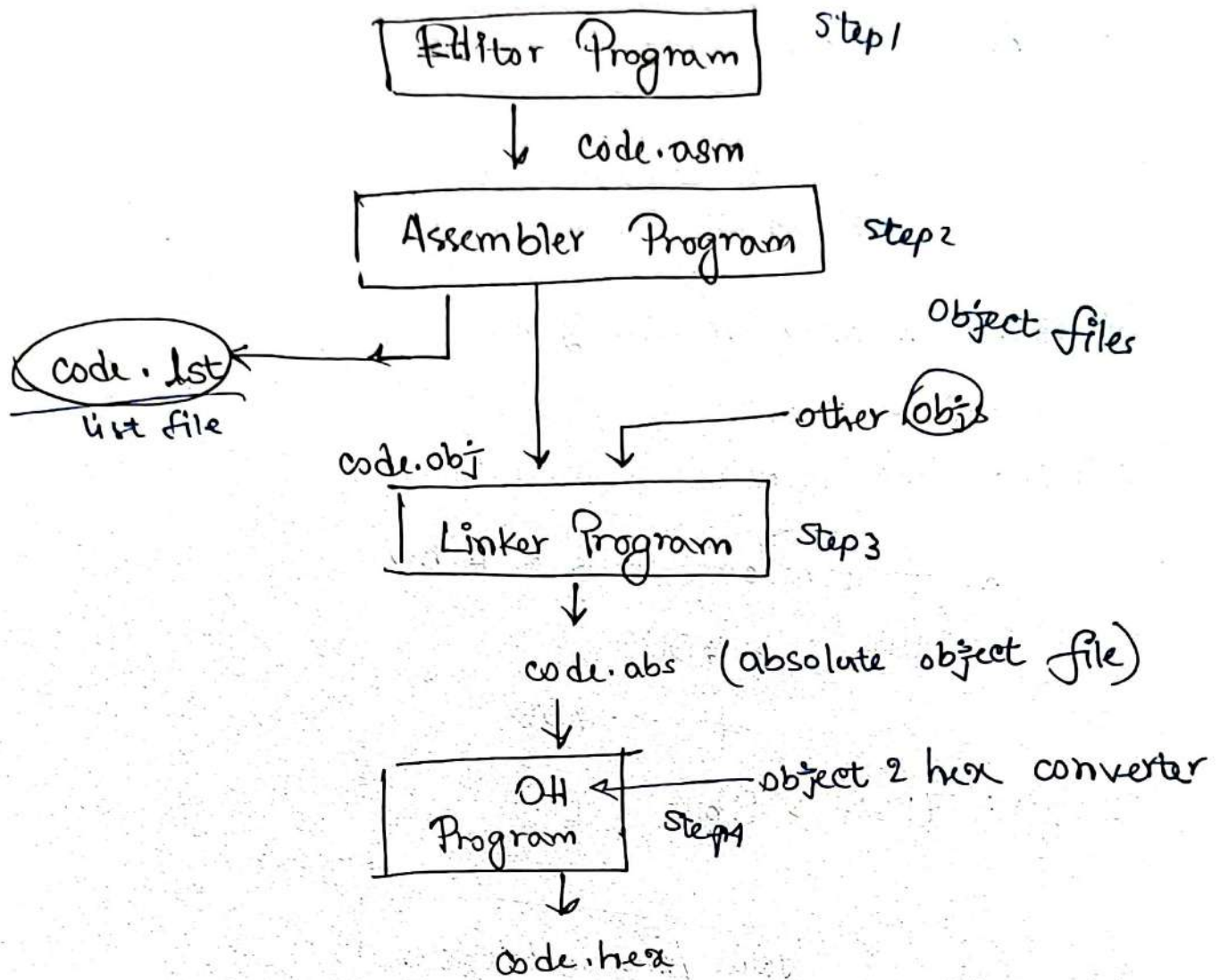
'special character'

Structure of Assembly Language

[label:] mnemonic [operands] [: comment]

Shayan

Running an ASM Program



List file की 21 उपयोग?

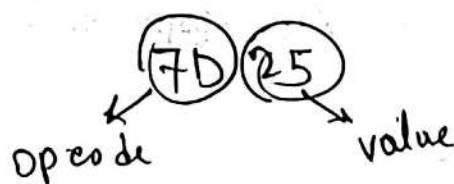
* Useful for programmers to detect syntax errors

Program Counter (16 bit) 0000 to FFFFH

'ORG' organization.

How ROM saves the code

Machine lang



an

Fields of Assembly Language

(11)

[label:] mnemonic [operands] [; comment]

DB → define data directive
↓
decimal, binary, hex, ASCII string
(use quotes)
↓
Byte
Define

Origin (ORG)

Equate (EQU)

#Define pi 3.1416

↳ count equ 30 ; count = 30

define constant without occupying memory storage

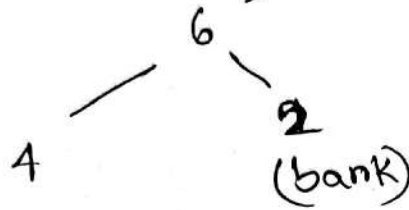
Pro

↳ 'Global Edit' final boss

↳ একতরফে change বাহা সমতরফে change হয়

Shayan

PSW (8bit)

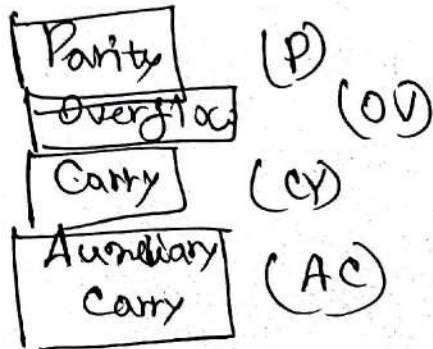


2 (user define)

1, 5 " - "

4 to flag (conditional)

RS0 ← PSW.3
RS1 ← PSW.4



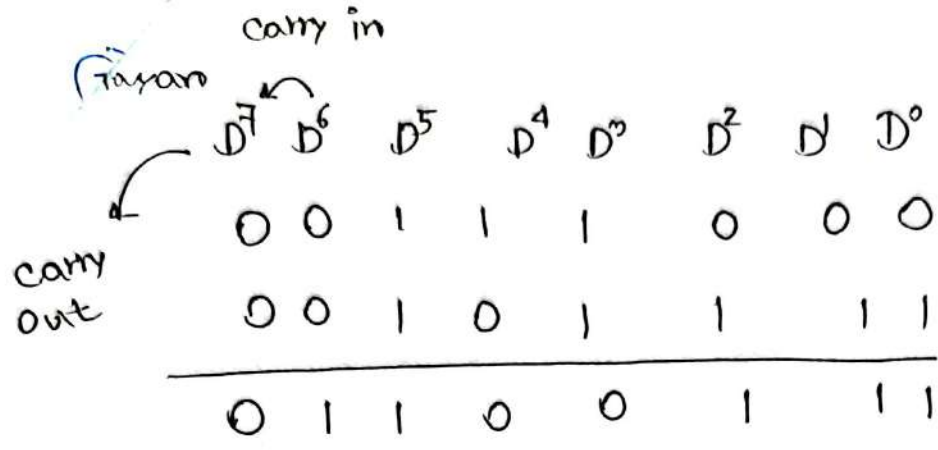
00-FF

Memory



Bank (Categories)

RS1	RS0	Register	Bank	Address
0	0			
0	1			
1	0			
1	1			



AC = 1
CY = 0
P = 1

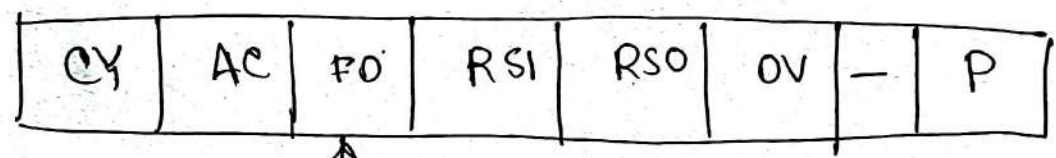
odd 1s? = 1
even 1s? = 0

if carry in = carry out [no overflow]

carry in $\neq$ carry out [overflow] OV = 0

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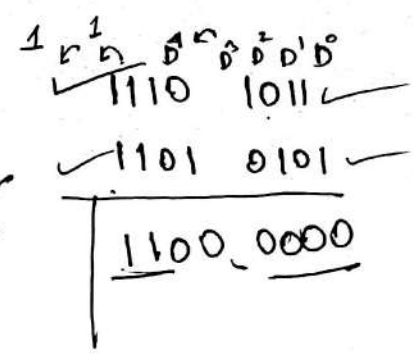
D3 $\rightarrow$ D4 Carry (STC)
Aux Carry 1



bank 0
by default

A B C D E
10 11 12 13 14

~~V.V.G~~
EB
D5



AC = 1
CY = 1
P = 0
OV = 0

Carry out $\rightarrow$ 1

Carry In 1